

1 Principles of Monte Carlo

1.1 Introduction Monte Carlo integration

Consider estimating $\alpha = \int_0^1 f(x)\,dx$. If $U \sim \text{Unif}(0,1)$, then $\alpha = \mathbb{E}[f(U)]$. Given independent samples $U_1,\ldots,U_n \sim \text{Unif}(0,1)$, the Monte Carlo estimator is

$\hat{\alpha}_n = \frac{1}{n} \sum_{i=1}^n f(U_i)$.

By the strong law of large numbers, $\hat{\alpha}_n \rightarrow \alpha$ a.s. as $n \rightarrow \infty$. By the central limit theorem,

$\hat{\alpha}_n - \alpha \approx N(0,\sigma_f^2/n)$,

where $\sigma_f^2 = \text{var}(f(U))$. Thus the standard error is $\sigma_f/\sqrt{n}$. Halving the standard error requires increasing the number of samples by a factor of 4.

Dimensionality

For one-dimensional integration, Monte Carlo is usually not competitive with deterministic methods such as the trapezoidal rule, whose error is $O(n^{-2})$. The advantage of Monte Carlo is that its convergence rate is not directly sensitive to the dimension of the problem, while many deterministic integration methods suffer from the curse of dimensionality.

1.2 First Example: European Call Option

Risk-neutral valuation

Let $S(t)$ be the stock price at time t . A European call option with strike K and maturity T has payoff $(S(T) - K)^+ = \max\{0,S(T) - K\}$. Under the risk-neutral measure $\mathbb{Q}$, the time-0 price is

$V_0 = \mathbb{E}_0^{\mathbb{Q}}\left[e^{-rT}(S(T) - K)^+\right]$,

where r is the continuously compounded risk-free rate.

Stock-price dynamics

Under $\mathbb{Q}$, geometric Brownian motion satisfies $\frac{dS(t)}{S(t)} = r\,dt + \sigma\,dW(t)$, where $W(t)$ is a standard Brownian motion. The exact solution at time T is

$S(T) = S(0)\exp\left((r - \frac{1}{2}\sigma^2)T + \sigma W(T)\right)$.

Since $W(T) = \sqrt{T}Z$ with $Z \sim N(0,1)$,

$S(T) = S(0)\exp\left((r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z\right)$.

Simulation algorithm

For $i = 1,\ldots,n$, generate $Z_i \sim N(0,1)$, then set

$S_i(T) = S(0)\exp\left((r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z_i\right)$,

$C_i = e^{-rT}(S_i(T) - K)^+$.

The Monte Carlo estimator is $\hat{C}_n = \frac{1}{n} \sum_{i=1}^n C_i$. It is unbiased, $\mathbb{E}[\hat{C}_n] = C$, and strongly consistent, $\hat{C}_n \rightarrow C$ with probability 1.

Confidence interval

The sample standard deviation of the simulated discounted payoffs is

$s_C = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (C_i - \hat{C}_n)^2}$.

An asymptotic $1 - \delta$ confidence interval is

$\hat{C}_n \pm z_{\delta/2} \frac{s_C}{\sqrt{n}}$.

Since s_C is estimated, one may replace the normal quantile with the corresponding t -quantile with $n - 1$ degrees of freedom.

Black-Scholes benchmark

The Black-Scholes value of a European call is

$BS(S,\sigma,T,r,K) = S\Phi(d_1) - e^{-rT}K\Phi(d_2)$,

where

$d_1 = \frac{\log(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$,

$d_2 = \frac{\log(S/K) + (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$.

1.3 Simulation Layers

Layered construction

Monte Carlo derivative pricing is built in layers:

$U_1 \rightarrow Z_1 \rightarrow S_1(T)$ or $\{S_1(t_j)\}_{j=1}^m \rightarrow C_1 \rightarrow \hat{C}_n, s_C$.

The lower layers transform simple random variables into asset prices or paths. The upper layers transform paths into discounted payoffs and then into statistical summaries.

1.4 Second Example: Asian Option

Path dependence

An Asian option is path dependent. A discrete-average Asian call has payoff $(\bar{S} - K)^+$, where

$\bar{S} = \frac{1}{m} \sum_{j=1}^m S(t_j)$,

and $0 = t_0 < t_1 < \cdots < t_m = T$. Its time-0 value is

$\mathbb{E}_0^{\mathbb{Q}}\left[e^{-rT}(\bar{S} - K)^+\right]$.

To estimate this value, simulate the path values $S(t_1),\ldots,S(t_m)$, compute the average $\bar{S}$, discount the payoff, and average over independent replications.

1.5 Efficiency of Simulation Estimators

Central limit theorem

Suppose $\hat{C}_n = \frac{1}{n} \sum_{i=1}^n C_i$, where C_i are i.i.d., $\mathbb{E}_0^{\mathbb{Q}}[C_i] = C$, and $\text{var}(C_i) = \sigma_C^2 < \infty$. Then

$\frac{\hat{C}_n - C}{\sigma_C/\sqrt{n}} \Rightarrow N(0,1)$,

or equivalently

$\sqrt{n}(\hat{C}_n - C) \Rightarrow N(0,\sigma_C^2)$.

Hence $\hat{C}_n - C \approx N(0,\sigma_C^2/n)$. For two unbiased estimators of the same quantity, the estimator with lower variance is preferred, other things being equal.

1.6 Bias

General principle

Variance reduction and faster computation are useful only if the estimator converges to the correct value. Some estimators are biased for every finite n , but become asymptotically unbiased as computational effort increases.

Ratio estimator

For i.i.d. samples (X_i,Y_i) , the estimator $\bar{X}/\bar{Y}$ for $\mathbb{E}[X]/\mathbb{E}[Y]$ is generally biased because

$\mathbb{E}\left[\frac{\bar{X}}{\bar{Y}}\right] \neq \frac{\mathbb{E}[X]}{\mathbb{E}[Y]}$.

However,

$\frac{\bar{Y}}{\bar{X}} \rightarrow \frac{\mathbb{E}[X]}{\mathbb{E}[Y]}$

with probability 1, and the normalized error is asymptotically normal. The bias becomes negligible as $n \rightarrow \infty$.

Model discretization error

If the exact geometric Brownian motion transition is replaced by the Euler recursion

$S((j+1)h) = S(jh) + rS(jh)h + \sigma S(jh)\sqrt{h}Z_{j+1}$,

then the simulated path distribution is not exactly the same as that implied by the continuous-time model. Option values estimated from this approximate dynamics are biased.

This model discretization bias can be reduced or eliminated by: (i) using the exact transition when available; (ii) decreasing the time step h , at the cost of more transitions per path.

Payoff discretization error

For a continuously averaged Asian option,

$\bar{S} = \frac{1}{T} \int_0^T S(u)\,du$.

Even exact simulation at a finite set of dates cannot compute the continuous average exactly. The payoff must be approximated, which creates payoff discretization error. This bias can be made small by using a smaller simulation time step, but computational cost increases.

Model discretization error comes from approximating the underlying dynamics. Payoff discretization error comes from approximating the payoff functional.

1.7 Mean Square Error

Bias-variance decomposition

For an estimator $\hat{\alpha}$ of α , the mean square error is

$\text{MSE}(\hat{\alpha}) = \mathbb{E}\left[(\hat{\alpha} - \alpha)^2\right]$.

It decomposes as

$\text{MSE}(\hat{\alpha}) = (\mathbb{E}[\hat{\alpha}] - \alpha)^2 + \mathbb{E}\left[(\hat{\alpha} - \mathbb{E}[\hat{\alpha}])^2\right]$,

or

$\text{MSE}(\hat{\alpha}) = \text{Bias}^2(\hat{\alpha}) + \text{Variance}(\hat{\alpha})$.

Thus MSE balances bias and variance.

2 Generating Random Numbers and Random Variables

2.1 Random Number Generation

Uniform inputs

Monte Carlo simulation is driven by apparently random numbers. Ideally we want $U_1,U_2,\ldots$ such that each $U_i \sim \text{Unif}[0,1]$ and the U_i are mutually independent. Independence is crucial: U_i should not be predictable from $U_1,\ldots,U_{i-1}$.

A pseudorandom number generator produces a finite deterministic sequence $u_1,\ldots,u_K \in [0,1]$. A good generator makes short segments hard to distinguish from independent uniforms. Desirable properties: long period, reproducibility, speed, portability, and statistical randomness.

Linear congruential generator

A linear congruential generator is

$x_{i+1} = ax_i \bmod m$, $u_{i+1} = x_{i+1}/m$.

Here a is the multiplier, m the modulus, and $x_0 \in \{1,\ldots,m-1\}$ the seed. The modulo operation is $y \bmod m = y - \lfloor y/m \rfloor m$.

Once a value repeats, the whole sequence repeats. The generator has full period if it produces all $m - 1$ nonzero values before repeating. Large m alone is not enough: poor choices of a,m can produce short cycles.

2.2 General Sampling Methods

Uniform distribution

Assume independent uniforms satisfying $P(U_i \leq u) = 0$ if $u < 0$, $P(U_i \leq u) = u$ if $0 \leq u \leq 1$, and $P(U_i \leq u) = 1$ if $u > 1$. Sampling methods transform these uniforms into variables with desired distributions.

2.3 Inverse Transform Method

General method

To sample X with CDF F , set

$X = F^{-1}(U)$, $U \sim \text{Unif}[0,1]$.

If F is not strictly increasing, use the generalized inverse

$F^{-1}(u) = \inf\{x : F(x) \geq u\}$.

This chooses the smallest compatible x . Validity follows from

$P(X \leq x) = P(F^{-1}(U) \leq x) = P(U \leq F(x)) = F(x)$.

Flat sections of F have zero probability: if F is constant on $[a,b]$, then $P(a < X \leq b) = F(b) - F(a) = 0$.

Examples

Exponential with mean θ : $F(x) = 1 - e^{-x/\theta}$ for $x \geq 0$, so

$X = -\theta \log(1 - U)$, equivalently $X = -\theta \log U$.

Rayleigh example:

$F(x) = 1 - e^{-2x(x-b)}$, $x \geq b$,

so the inverse-transform sample is

$X = \frac{b}{2} + \frac{\sqrt{b^2 - 2\log U}}{2}$.

Discrete distribution: for outcomes $c_1 < \cdots < c_n$ with probabilities p_i , define $q_i = 0$ and

$q_i = \sum_{j=1}^i p_j$.

Generate U , find K with $q_{K-1} < U \leq q_K$, and set $X = c_K$. Binary search can implement the lookup.

Conditional sampling

To sample X conditional on $a < X \leq b$, with $F(a) < F(b)$, generate

$V = F(a) + [F(b) - F(a)]U$.

Then $F^{-1}(V)$ has the conditional distribution because

$P(F^{-1}(V) \leq x) = \frac{F(x) - F(a)}{F(b) - F(a)}$.

2.4 Acceptance-Rejection Method

General method

To sample from a target density f on $\mathcal{X}$, choose a convenient proposal density g and a constant c such that

$f(x) \leq c g(x)$, $x \in \mathcal{X}$.

Generate $X \sim g$ and $U \sim \text{Unif}[0,1]$. Accept X if

$U \leq \frac{f(X)}{cg(X)}$;

otherwise reject and repeat. If Y is the accepted value, then $P(Y \in A) = P(X \in A \mid U \leq f(X)/(cg(X)))$.

The acceptance probability is

$P(U \leq f(X)/(cg(X))) = \int_{\mathcal{X}} \frac{f(x)}{cg(x)} g(x)dx = 1/c$.

Hence

$P(Y \in A) = c \int_A \frac{f(x)}{cg(x)} g(x)dx = \int_A f(x)dx$.

Thus Y has density f . The number of proposals until acceptance is geometric with mean c , so smaller c and tighter bounds are better.

Beta distribution

The beta density on $[0,1]$ is

$f(x) = \frac{\Gamma(\alpha_1 + \alpha_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)} x^{\alpha_1-1} (1-x)^{\alpha_2-1}$.

Here

$B(\alpha_1,\alpha_2) = \int_0^1 x^{\alpha_1-1} (1-x)^{\alpha_2-1} dx = \frac{\Gamma(\alpha_1)\Gamma(\alpha_2)}{\Gamma(\alpha_1 + \alpha_2)}$,

and

$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx$, $\text{Re}(z) > 0$.

For integer $n > 0$, $\Gamma(n) = (n-1)!$. If $\alpha_1,\alpha_2 \geq 1$ and at least one exceeds 1, the mode is $\frac{\alpha_1-1}{\alpha_1+\alpha_2-2}$.

Let c be the maximum value of f and use $g(x) = 1$ on $[0,1]$. Generate U_1,U_2 until $cU_2 \leq f(U_1)$, then return U_1 .

Normal from double exponential

For the double exponential density $g(x) = e^{-|x|}/2$ and standard normal density $f(x) = \frac{1}{\sqrt{2\pi}} \exp(-x^2/2)$,

the ratio satisfies

$\frac{f(x)}{g(x)} = \sqrt{\frac{\pi}{2}} e^{-x^2/2+|x|} \leq \sqrt{\frac{\pi}{2}} \approx 1.3155 \approx c$.

The rejection test may be written as

$u > \exp\left(-\frac{1}{2}x^2 + |x| - \frac{1}{2}\right) = \exp\left(-\frac{1}{2}(|x| - 1)^2\right)$.

Using symmetry: generate U_1,U_2,U_3 , set $X = -\log U_1$, reject if $U_2 > \exp[-0.5(X - 1)^2]$, then set $X \leftarrow -X$ if $U_3 \leq 0.5$.

2.5 Normal Random Variables and Vectors

Multivariate normal

A d -dimensional normal is written $N(\mu,\Sigma)$, where μ is a d -vector and Σ is a $d \times d$ covariance matrix. A covariance matrix must be symmetric and positive semidefinite:

$x^\top \Sigma x \geq 0$, $x \in \mathbb{R}^d$.

If Σ is positive definite, the density is

$\phi_{\mu,\Sigma}(x) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\left[-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right]$.

If $X \sim N(\mu,\Sigma)$, then $X_i \sim N(\mu_i,\sigma_i^2)$ with $\sigma_i^2 = \Sigma_{ii}$, and

$\text{Cov}(X_i,X_j) = \mathbb{E}[(X_i - \mu_i)(X_j - \mu_j)] = \Sigma_{ij}$,

$\rho_{ij} = \frac{\Sigma_{ij}}{\sigma_i\sigma_j}$, equivalently $\Sigma_{ij} = \sigma_i\sigma_j\rho_{ij}$.

If Σ is positive semidefinite but singular, define $X = \mu + AZ$ with $Z \sim N(0,I_d)$ and $AA^\top = \Sigma$. Then X has no full d -dimensional density when $\text{rank}(\Sigma) < d$.

Useful properties

Linear transformation: if $X \sim N(\mu,\Sigma)$, then

$AX \sim N(A\mu, A\Sigma A^\top)$.

Conditioning formula: for a partitioned normal vector with $\Sigma_{[22]}$ full rank,

$(X_{[1]} \mid X_{[2]} = x) \sim N(\mu_{[1]} + \Sigma_{[12]}\Sigma_{[22]}^{-1}(x - \mu_{[2]}), \Sigma_{[11]} - \Sigma_{[12]}\Sigma_{[22]}^{-1}\Sigma_{[21])}$.

Moment generating function:

$\mathbb{E}[\exp(\theta^\top X)] = \exp(\mu^\top \theta + \frac{1}{2}\theta^\top \Sigma \theta)$.

2.6 Generating Normals

Box-Muller method

The Box-Muller method generates a bivariate standard normal. If $Z \sim N(0,I_2)$, then $R = Z_1^2 + Z_2^2$ is exponential with mean 2:

$P(R \leq s) = 1 - e^{-s/2}$.

Given R,Z_1,Z_2 , Z is uniformly distributed on the circle of radius $\sqrt{R}$. Generate

independent U_1,U_2 , set

$R = -2\log U_1$, $V = 2\pi U_2$,

and return

$Z_1 = \sqrt{R}\cos V$, $Z_2 = \sqrt{R}\sin V$.

Multivariate normals

To sample $X \sim N(\mu,\Sigma)$, generate $Z \sim N(0,I_d)$, find A such that

$AA^\top = \Sigma$, and set $X = \mu + AZ$. Then $X \sim N(\mu,\Sigma)$.

2.7 Cholesky Factorization

Positive definite case

Cholesky represents $\Sigma = AA^\top$ with A lower triangular. For

$\Sigma = \begin{pmatrix} \sigma_1^2 & & \\ \sigma_1\sigma_2\rho & \sigma_2^2\rho^2 & \\ \sigma_1\sigma_2\rho & \sigma_2^2\rho^2 & \sigma_2^2 \end{pmatrix}$,

a Cholesky factor is

$A = \begin{pmatrix} \sigma_1 & & 0 \\ \rho\sigma_2 & \sqrt{1-\rho^2}\sigma_2 & \end{pmatrix}$.

Thus

$X_1 = \mu_1 + \sigma_1 Z_1$,

$X_2 = \mu_2 + \sigma_2\rho Z_1 + \sigma_2\sqrt{1-\rho^2}Z_2$.

For general d , use

$\Sigma_{ij} = \sum_{k=1}^i A_{ik}A_{jk}$, $j \leq i$.

The recursive formulas are

$A_{ij} = \frac{\Sigma_{ij} - \sum_{k=1}^{j-1} A_{ik}A_{jk}}{A_{jj}}$,

3.6 Multi-Asset GBM

Correlated assets
Let X_t be standard Brownian motions with correlations ρ_{ij} . A multidimensional GBM is

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + \sigma_i dX_i(t), \quad i = 1, \dots, d.$$

Define $\Sigma_{ij} = \sigma_i \sigma_j \rho_{ij}$. Then $(\sigma_1 X_1, \dots, \sigma_d X_d) \sim \text{BM}(0, \Sigma)$ and $S = (S_1, \dots, S_d) \sim \text{GBM}(\mu, \Sigma)$.
The price covariance is

$$\text{Cov}[S_i(t), S_j(t)] = S_i(0)S_j(0)e^{(\mu_i + \mu_j)t} \left(\rho_{ij} \sigma_i \sigma_j t - 1 \right).$$

If $AA^\top = \Sigma$ and a_i is the i th row of A ,

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + a_i dW(t) = \mu_i dt + \sum_{j=1}^d A_{ij} dW_j(t).$$

Simulation and payoffs

With independent $Z_k = (Z_{k1}, \dots, Z_{kd}) \sim N(0, I)$,

$$S_i(t_{k+1}) = S_i(t_k) \exp \left(\left(\mu_i - \frac{1}{2} \sigma_i^2 \right) \Delta t_k + \sqrt{\Delta t_k} \sum_{j=1}^d A_{ij} Z_{k+1,j} \right).$$

Under the risk-neutral measure, $\mu_i = r - \delta_i$. Typical multi-asset payoffs:

Spread option:

$$[|S_1(T) - S_2(T)| - K]^+.$$

Basket option:

$$([c_1 S_1(T) + c_2 S_2(T) + \dots + c_d S_d(T)] - K)^+.$$

Outperformance option:

$$(\max(c_1 S_1(T), c_2 S_2(T), \dots, c_d S_d(T)) - K)^+.$$

Two-asset barrier example:

$$\mathbf{1}_{[\min_{i=1,\dots,n} S_2(t_i) < b] - (K - S_1(T))^+}.$$

4 Variance Reduction Techniques

4.1 Control Variates

Basic method
Let $Y_1, \dots, Y_r$ be i.i.d. outputs for estimating $\mathbb{E}[Y]$, and suppose X_i is generated with Y_i and $\mathbb{E}[X]$ is known. For fixed b define $\bar{Y}_i(b) = Y_i - b(X_i - \mathbb{E}[X])$, and estimate $\bar{Y}(b) = \bar{Y} - b(\bar{X} - \mathbb{E}[X]) = \frac{1}{n} \sum_{i=1}^n (Y_i - b(X_i - \mathbb{E}[X]))$. This is unbiased for all b . Its per-replication variance is $\sigma^2(b) = \text{var}[Y_i(b)] = \sigma_Y^2 - 2b\sigma_X \sigma_Y \rho_{XY} + b^2 \sigma_X^2$. The optimal coefficient is

$$b^* = \frac{\sigma_Y}{\sigma_X} \rho_{XY} = \frac{\text{Cov}(X, Y)}{\text{var}(X)}.$$

The resulting variance ratio relative to plain Monte Carlo is

$$\frac{\text{var}[\bar{Y} - b^*(\bar{X} - \mathbb{E}[X])]}{\text{var}[\bar{Y}]} = 1 - \rho_{XY}^2.$$

This speed-up is approximately $1/(1 - \rho_{XY}^2)$. The sign of ρ_{XY} is absorbed by b^* ; only $|\rho_{XY}|$ matters.

Estimated coefficient and CI

In practice use the sample regression coefficient

$$\hat{b}_n = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

Then $\hat{b}_n \rightarrow b^*$ a.s. and $\bar{Y}(\hat{b}_n) = \frac{1}{n} \sum_{i=1}^n \{Y_i - \hat{b}_n(X_i - \mathbb{E}[X])\}$.

Using $\hat{b}_n$ creates finite-sample bias, but has the same asymptotic precision as using b^* . For fixed b ,

$$\frac{\bar{Y}(b) - \mathbb{E}[\bar{Y}]}{\sigma(b)/\sqrt{n}} \Rightarrow N(0, 1),$$

with sample standard deviation

$$s(b) = \left[\frac{1}{n-1} \sum_{i=1}^n (Y_i(b) - \bar{Y}(b))^2 \right]^{1/2}.$$

Hence an asymptotic CI is $\bar{Y}(b) \pm z_{\delta/2} s(b)/\sqrt{n}$. Also

$$\frac{\bar{Y}(\hat{b}_n) - \mathbb{E}[\bar{Y}]}{s(\hat{b}_n)/\sqrt{n}} \Rightarrow N(0, 1).$$

Financial controls

Any martingale with known initial value is a natural control. Under risk-neutral pricing with constant r , $e^{-rt}S(t)$ is a martingale, so $\mathbb{E}[e^{-rt}S(T)] = S(0)$. For discounted payoff Y ,

$$\frac{1}{n} \sum_{i=1}^n \{Y_i - b[e^{-rt}S(T) - S(0)]\}$$

uses the stock as control. For a call, this works best near $K = 0$ and may weaken for deep OTM strikes. A stronger control can be a tractable option. For Asian options,

$$\bar{S}_A = \frac{1}{n} \sum_{i=1}^n S(t_i), \quad \bar{S}_G = (\prod_{i=1}^n S(t_i))^{1/n}.$$

Under GBM, $\bar{S}_G$ is lognormal and the geometric-averaged Asian option can be priced analytically; its payoff is often highly correlated with the arithmetic-averaged Asian payoff.

4.2 Antithetic Variates

Estimator

Antithetic variates pair inputs to create negative dependence while preserving the marginal law. If $U \sim \text{Unif}(0, 1]$, use U and $1 - U$; for a symmetric normal input, use Z and $-Z$.

Let $(Y_i, \bar{Y}_i)$ be i.i.d. pairs with the same marginal distribution. The estimator is

$$\bar{Y}_{AV} = \frac{1}{n} \sum_{i=1}^n Y_i + \sum_{i=1}^n \bar{Y}_i = \frac{1}{n} \sum_{i=1}^n \frac{Y_i + \bar{Y}_i}{2}.$$

Apply the CLT to pair averages:

$$\frac{\bar{Y}_{AV} - \mathbb{E}[\bar{Y}]}{\sigma_{AV}/\sqrt{n}} \Rightarrow N(0, 1), \quad \sigma_{AV}^2 = \text{var} \left(\frac{Y_1 + \bar{Y}_1}{2} \right).$$

A CI is $\bar{Y}_{AV} \pm z_{\delta/2} s_{AV}/\sqrt{n}$, where s_{AV} is the sample sd of pair averages.

When it helps

Compare n antithetic pairs with $2n$ independent samples. Since

$$\text{var}[Y_i + \bar{Y}_i] = 2\text{var}[Y_i] + 2\text{Cov}(Y_i, \bar{Y}_i),$$

antithetic sampling reduces variance iff $\text{Cov}(Y_i, \bar{Y}_i) < 0$. Monotone outputs often satisfy this when paired by $U, 1 - U$ or $Z, -Z$.

GBM call example

For a call, pair Z_i and $-Z_i$:

$$S_i^+ = S_0 \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) T + \sigma \sqrt{T} Z_i \right),$$

$$S_i^- = S_0 \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) T - \sigma \sqrt{T} Z_i \right).$$

Set $X_i = e^{-rT} (S_i^+ - K)^+$ and $\bar{Y}_i = e^{-rT} (S_i^- - K)^+$. Then

$$\hat{v} = \frac{1}{n} \sum_{i=1}^n \frac{X_i + \bar{Y}_i}{2},$$

$$SE = \left[\frac{1}{n(n-1)} \left\{ \sum_{i=1}^n \left(\frac{X_i + \bar{Y}_i}{2} \right)^2 - n\hat{v}^2 \right\} \right]^{1/2}.$$

4.3 Stratified Sampling

Basic method

Let $A_1, \dots, A_K$ be disjoint strata with probabilities p_i . Then

$$\mathbb{E}[Y] = \sum_{i=1}^K p_i \mathbb{E}[Y | Y \in A_i].$$

With proportional allocation $n_i = np_i$ and conditional draws $Y_{ij} \sim (Y | Y \in A_i)$,

$$\hat{Y} = \sum_{i=1}^K p_i \frac{1}{n_i} \sum_{j=1}^{n_i} Y_{ij} = \frac{1}{n} \sum_{i=1}^K p_i \sum_{j=1}^{n_i} Y_{ij}.$$

Stratification removes the random variation in the number of observations landing in each stratum.

General allocation

Often strata are defined by another variable X . If A_i partition the support of X ,

$$\mathbb{E}[Y] = \sum_{i=1}^K p_i \mathbb{E}[Y | X \in A_i].$$

For $n_i \geq 1$, $\sum_i n_i = n$, and $q_i = n_i/n$,

$$\hat{Y} = \sum_{i=1}^K p_i \frac{q_i}{n_i} \sum_{j=1}^{n_i} Y_{ij} = \frac{1}{n} \sum_{i=1}^K p_i \frac{q_i}{q_i} \sum_{j=1}^{n_i} Y_{ij}.$$

Let $\mu_i = \mathbb{E}[Y | X \in A_i]$ and $\sigma_i^2 = \text{var}(Y | X \in A_i)$. Then $\hat{Y}$ is unbiased and

$$\text{var}[\hat{Y}] = \sum_{i=1}^K p_i \frac{q_i^2}{n_i} \frac{\sigma_i^2}{q_i} = \frac{\sigma^2(q)}{n}, \quad \sigma^2(q) = \sum_{i=1}^K p_i \frac{q_i^2}{q_i} \sigma_i^2.$$

With fixed K , $\sqrt{n}(\hat{Y} - \mu) \Rightarrow N(0, \sigma^2(q))$, and

$$s^2(q) = \sum_{i=1}^K p_i \frac{q_i^2}{q_i} s_i^2$$

estimates $\sigma^2(q)$.

Uniform and inverse-transform strata

For $U \sim \text{Unif}(0, 1]$, equal strata are $A_i = ((i-1)/n, i/n]$. With $U_i \sim \text{Unif}(0, 1]$,

$$V_i = \frac{i-1}{n} + \frac{U_i}{n}, \quad \bar{Y} = \frac{1}{n} \sum_{i=1}^n \bar{Y}(V_i).$$

For a distribution with CDF F , choose probabilities p_i and quantiles $a_i = F^{-1}(p_1 + \dots + p_i)$. Conditional simulation in stratum $(a_{i-1}, a_i]$ uses

$$V = F(a_{i-1} - 1) + U \{F(a_i) - F(a_{i-1})\}, \quad F^{-1}(V) \sim (Y | Y \in A_i).$$

Proportional and optimal allocation

For proportional allocation $q_i = p_i$,

$$\sigma^2(q) = \sum_{i=1}^K p_i \sigma_i^2.$$

Plain variance decomposes as

$$\text{var}(Y) = \sum_{i=1}^K p_i \sigma_i^2 + \sum_{i=1}^K p_i \mu_i^2 - \left(\sum_{i=1}^K p_i \mu_i \right)^2.$$

By Jensen, proportional stratification cannot increase variance, with strict gain unless all μ_i are equal. Optimizing $\sigma^2(q)$ gives

$$q_i^* = \frac{p_i \sigma_i}{\sum_{i=1}^K p_i \sigma_i}, \quad \sigma^2(q^*) = \left(\sum_{i=1}^K p_i \sigma_i \right)^2.$$

Thus allocate samples proportional to stratum probability times within-stratum sd.

Call option strata

For a GBM call, write $W_T = \sqrt{T} \Phi^{-1}(U)$ with $U \sim \text{Unif}(0, 1]$ and

$$h(u) = \left(S_0 \exp \left\{ -\frac{1}{2} \sigma^2 T + \sigma \sqrt{T} \Phi^{-1}(u) \right\} - e^{-rT} K \right)^+.$$

With k equal strata, $U_{ij} = (i-1)/k + V_{ij}/k$, $\hat{\mu}_i = n_i^{-1} \sum_{j=1}^{n_i} h(U_{ij})$, and

$$\hat{\mu} = \frac{1}{k} \sum_{i=1}^k \hat{\mu}_i, \quad SE = \frac{1}{k} \left(\sum_{i=1}^k \frac{s_i^2}{n_i} \right)^{1/2}.$$

4.4 Importance Sampling

Change of measure

To estimate $\mu = \mathbb{E}[h(X)]$ with density f , choose an alternative density g such that $g(x) = 0$ implies $f(x) = 0$. If $Y \sim g$,

$$\mu = \int h(x) f(x) dx = \int h(x) \frac{f(x)}{g(x)} g(x) dx = \mathbb{E}_g \left[h(Y) \frac{f(Y)}{g(Y)} \right].$$

In a discrete model, use $h(Y_i)/p(Y_i)/\bar{p}(Y_i)$.

Variance and zero-variance density

The IS estimator has variance

$$\frac{1}{n} \text{var}_g \left[h(Y) \frac{f(Y)}{g(Y)} \right].$$

For $h \geq 0$, the zero-variance density is

$$g^*(x) = c h(x) f(x), \quad \frac{1}{c} = \int h(x) f(x) dx = \mu.$$

It is unusable directly because it requires μ , but it says g should mimic $h(x)f(x)$.

Normal mean shift and options

For $X \sim N(0, 1)$, choose $Y \sim N(\theta, 1)$. The likelihood ratio is

$$\frac{f(X)}{g(X)} = \exp \left\{ -\theta x + \frac{1}{2} \theta^2 \right\}.$$

Mode matching sets $\theta = \arg \max_x h(x) f(x)$. For a binary call,

$$\mathbb{E}[e^{-rT} \mathbf{1}_{\{S_T \geq K\}}] = \mathbb{E}[e^{-rT} \mathbf{1}_{\{X \geq b\}}],$$

where

$$b = \frac{\log(K/S_0) - (r - \sigma^2/2)T}{\sigma \sqrt{T}}.$$

Here $h(x) = e^{-rT} \mathbf{1}_{\{x \geq b\}}$, $x^* = \max\{b, 0\}$, and with $Y \sim N(x^*, 1)$ the output is

$$h(Y) \frac{f(Y)}{g(Y)} = \begin{cases} 0, & Y < b, \\ \exp\{-rT - x^* Y + (x^*)^2/2\}, & Y \geq b. \end{cases}$$

For a vanilla call,

$$h(x) = \left(S_0 \exp \left\{ -\frac{1}{2} \sigma^2 T + \sigma \sqrt{T} x \right\} - e^{-rT} K \right)^+,$$

and the shift pushes sampling toward exercise states.

Exponential tilting

For density f , the exponential tilt family is

$$f_\theta(x) = \frac{e^{\theta x} f(x)}{\int_{-\infty}^{\infty} e^{\theta x} f(x) dx}, \quad \theta \in \mathbb{R}.$$

For a pmf p ,

$$p_\theta(x) = \frac{e^{\theta x} p(x)}{\sum_{i=1}^n e^{\theta x_i} p(x_i)}.$$

For normal X , this is mean shifting. For rare large losses, take $\theta > 0$.

Bernoulli rare event

Let independent defaults satisfy $P(X_k = 1) = p_k$ and $L = \sum_{k=1}^m X_k$. To estimate

$P(L > x)$ with $x \gg \mathbb{E}[L]$, tilt each Bernoulli:

$$\bar{p}_k = P(\bar{Y}_k = 1) = \frac{p_k e^\theta}{1 + p_k(e^\theta - 1)}, \quad \bar{L} = \sum_{k=1}^m \bar{Y}_k.$$

The IS output is

$$H = \mathbb{I}_{\{L > x\}} \prod_{k=1}^m \left(\frac{\bar{p}_k}{p_k} \right)^{\bar{Y}_k} \left(\frac{1 - \bar{p}_k}{1 - p_k} \right)^{1 - \bar{Y}_k}.$$

With

$$\phi(\theta) = \sum_{k=1}^m \log[1 + p_k(e^\theta - 1)],$$

one has $E[H^2] \leq e^{-2\phi(\theta) + 2\phi(\theta)}$. Choose $\theta > 0$ by solving $\phi'(\theta) = x$.

4.5 Cross-Entropy Method

Principle

Cross-entropy selects a parametric IS density f_θ close to the zero-variance density

$$g^*(x) = \frac{1}{\mu} h(x) f(x).$$

Minimize

$$R(g^* || f_\theta) = \int \log \left(\frac{g^*(x)}{f_\theta(x)} \right) g^*(x) dx,$$

equivalently maximize

$$\int h(x) f(x) \log f_\theta(x) dx.$$

The population first-order equation is

$$\mathbb{E} \left[h(X) \frac{\partial}{\partial \theta} \log f_\theta(X) \right] = 0.$$

Basic algorithm

Using pilot samples $X_1, \dots, X_N \sim f$, solve

$$\frac{1}{N} \sum_{k=1}^N h(X_k) \frac{\partial}{\partial \theta} \log f_\theta(X_k) = 0.$$

For $f = N(0, I_m)$ and $f_\theta = N(\theta, I_m)$,

$$\hat{\theta} = \frac{\sum_{k=1}^N h(X_k) X_k}{\sum_{k=1}^N h(X_k)}.$$

Then sample $Y_i \sim f_{\hat{\theta}}$ and average

$$H_i = h(Y_i) \frac{f(Y_i)}{f_{\hat{\theta}}(Y_i)}.$$

For a one-dimensional normal shift, $H_i = h(Y_i) \exp\{-\hat{\theta} Y_i + \hat{\theta}^2/2\}$. For very rare events, the basic method may fail because $\sum_k h(X_k) + 0$ in the pilot run.

Iterative cross-entropy

Let $g_\theta(x) = f(x)/f_\theta(x)$. If pilot samples are drawn from f_v , then maximize $\mathbb{E}_v[h(Y)f_v(Y) \log f_\theta(Y)]$, and solve

$$\mathbb{E}_v \left[h(Y) f_v(Y) \frac{\partial}{\partial \theta} \log f_\theta(Y) \right] = 0.$$

At iteration j , sample $Y_k \sim f_{\hat{\theta}_j}$ and update from

$$\frac{1}{N} \sum_{k=1}^N h(Y_k) \ell_{\hat{\theta}_j}(Y_k) \frac{\partial}{\partial \theta} \log f_{\hat{\theta}_j}(Y_k) = 0.$$

For $N(\theta, I_m)$,

$$\hat{\theta}_{j+1} = \frac{\sum_{k=1}^N h(Y_k) \ell_{\hat{\theta}_j}(Y_k) Y_k}{\sum_{k=1}^N h(Y_k) \ell_{\hat{\theta}_j}(Y_k)}.$$

Here

$$\ell_{\hat{\theta}_j}(Y_k) = \exp(-(\hat{\theta}_j \cdot Y_k) + \|\hat{\theta}_j\|^2/2).$$

Usually only a few iterations are needed; convergence means the updates oscillate in a small range rather than become deterministic.

5 Mean-variance and Optimization

5.1 Portfolio Constraints

Review: Markowitz Mean-Variance Problem

Recall the standard Markowitz Mean-Variance Problem:

$$\begin{aligned} \min_{w_0, w} \quad & \frac{1}{2} w^\top C w \\ \text{s.t.} \quad & w_0 r_f + w^\top \mathbf{1} = z \\ & w_0 + w^\top \mathbf{1}_n = 1 \end{aligned}$$

where w_0 is the weight in the risk-free asset, w is the vector of weights in risky assets, C is the covariance matrix, and μ is the expected return vector.

Equivalent formulation:

$$\begin{aligned} \min_w \quad & \frac{1}{2} w^\top C w \\ \text{s.t.} \quad & w^\top (\mu - r_f \mathbf{1}_n) = z - r_f \end{aligned}$$

The above formulation does not take into accounts of constraints in real trading.

Basic Trading Constraints

No short selling: $w_i \geq 0$, for $i = 1, \dots, n$.

No borrowing: $w_0 \geq 0$, which implies $\sum_{i=1}^n w_i \leq 1$.

Margin Requirement for Short Sale

Suppose that for stock i , the short sale of account of this stock should have $(1 + m_i)$ of the value of the short sale. Suppose the margin account earns a deterministic net return r_c . Denote by v_i the amount of margin prepared for the short sale of stock i , and denote $v := (v_1, \dots, v_n)$.

Portfolio Net Return:

$$R_{(w,v)} = w^\top R + v^\top r_c \mathbf{1}_n + (1 - w^\top \mathbf{1}_n - v^\top \mathbf{1}_n) r_f$$

$$= r_f + w^\top (R - r_f \mathbf{1}_n) + v^\top (r_c - r_f) \mathbf{1}_n$$

Moments: Variance is given by $\text{Var}(R_{(w,v)}) = w^\top C w$. Expectation is given by

$$\$$

Equivalent Utility Functions

Preferences remain the same if we replace $u(x)$ with $au(x) + b$ where $a > 0$. Thus, $v(x) := au(x) + b$ is equivalent to $u(x)$.

Examples of Utility Functions

Exponential utility ($\gamma > 0$):

$$u(x) = 1 - e^{-\gamma x}, \quad x \in \mathbb{R}$$

Power utility ($\gamma > 0, \gamma \neq 1$):

$$u(x) = x^{1-\gamma} / (1-\gamma), \quad x > 0$$

Logarithmic utility:

$$u(x) = \ln x, \quad x > 0$$

Quadratic utility ($\gamma > 0$):

$$u(x) = x - \frac{\gamma}{2} x^2, \quad x \in \mathbb{R}$$

6.8 Elicit Utility Function

Certainty Equivalent Method

Select reference points A and B . Propose lottery $(A, p; B, 1-p)$. Ask for certainty equivalent c . Then:

$$u(c) = pu(A) + (1-p)u(B)$$

$u(A)$ and $u(B)$ are set exogenously due to scaling freedom. Varying p maps the curve.

Probability Equivalent Method

Select reference points A and B . For a value x , ask for the probability p such that the individual accepts the lottery in place of x . Then:

$$u(x) = pu(A) + (1-p)u(B)$$

Tradeoff Method

Compare lotteries $(X, p; R)$ and $(x, p; R)$ with reference outcomes $R > r$. Vary X until indifference. Then compare $(Y, p; r)$ and $(y, p; R)$. By EUT:

$$u(X) - u(x) = u(Y) - u(y)$$

This establishes an indifference relation. Starting from a base x_0 , one can identify a sequence where $u(x_i) = ju(x_0)$.

6.9 Portfolio Selection with EU in Single Period

Consider a single period problem with initial wealth x_0 , assets with returns R_i , and portfolio weights w_i . Portfolio return is $R_w = \sum_{i=1}^n w_i R_i$. Terminal wealth is $x_0(1 + R_w)$. The objective is:

$$\max_{w_i} \mathbb{E}[u(x_0(1 + R_w))] \quad \text{s.t.} \quad \sum_{i=1}^n w_i = 1$$

Relation with Mean-Variance Criterion

For quadratic utility $u(x) = x - (\gamma/2)x^2$, the expected utility depends on $\mathbb{E}[R_w]$ and $\text{Var}[R_w]$. Thus, the optimal portfolio lies in the mean-variance set. Expected utility of terminal wealth:

$$x_0 \left\{ \mathbb{E}[1 + R_w] - (\gamma x_0/2) \mathbb{E}[(1 + R_w)^2] \right\} = x_0 \left\{ 1 - (\gamma x_0/2) + (1 - \gamma x_0) \mathbb{E}[R_w] + (\gamma x_0/2) (\mathbb{E}[R_w])^2 - (\gamma x_0/2) \text{Var}[R_w] \right\}$$

For exponential utility $u(x) = -e^{-\gamma x}$ and normally distributed returns, the expected utility is:

$$-\exp \left\{ -\gamma x_0 (1 + \mathbb{E}[R_w]) + \frac{\gamma^2 x_0^2}{2} \text{Var}[R_w] \right\}$$

Thus, the optimal portfolio can be on the mean-variance efficient frontier.

General Case: EU and mean-variance criteria differ. Using the Lagrange dual method, the optimal portfolio w^* satisfies:

$$\frac{\partial \mathbb{E}[u(x_0(1 + R_w))] }{\partial w_i} = \lambda, \quad i = 1, \dots, n \quad \text{s.t.} \quad \sum_{i=1}^n w_i = 1$$

7 CAPM and Arbitrage Pricing Theory

7.1 Market Equilibrium and CAPM model

Market equilibrium: at equilibrium price All investors have achieved their goals. Market is clear, demand=supply. **Capital asset pricing model (CAPM):** an equilibrium pricing model in framework of mean-variance analysis.

Assumptions

Each investor is a price-taker and mean-variance optimizer (although with different risk appetite). Everyone agrees on the probabilistic structure of assets (mean, covariance matrix of future asset prices). There is a unique risk-free rate r_f of borrowing and lending that is available to all. Market is frictionless: assets can be traded in any fraction and there is no transaction fee and other fees. The equilibrium price exists.

Market Portfolio Suppose there are n risky assets in the market with number of shares $N_i, i = 1, \dots, n$. The current prices are $P_i, i = 1, \dots, n$.

Total capital: $\sum_{i=1}^n N_i P_i$
Capitalization weight: $\tilde{w}_M = (\tilde{w}_M, 1, \dots, \tilde{w}_M, n)$
 $\tilde{w}_{M,i} = \frac{\text{capital asset } i}{\text{total capital}} = \frac{N_i P_i}{\sum_{i=1}^n N_i P_i}, i = 1, \dots, n$

Market portfolio: portfolio with capitalization weight.

Market Portfolio and Tangent Portfolio Each investor holds his optimal portfolio which is a mixture of risk-free asset and mutual tangent portfolio. The total assets held by all investors must be the mixture of risk-free asset and tangent portfolio. Under equilibrium, total holding of assets equals total supply. **This tangent portfolio is market portfolio.**

Suppose there are j investors in the market. Investor j has initial wealth x_j and invests in the portfolio $(w_{j0}, (1 - w_{j0})\tilde{w}_M)$, $j = 1, \dots, j$. The number of shares of the i -th risky asset investor j holds is $x_{j,i}(1 - w_{j0})\tilde{w}_{M,i}/P_i, i = 1, \dots, n$.

In equilibrium:

$$(\sum_{j=1}^j x_{j,i}(1 - w_{j0}))\tilde{w}_{M,i}/P_i = N_i$$

Thus, $\sum_{j=1}^j x_{j,i}(1 - w_{j0}) = \sum_{i=1}^n N_i P_i$, and for $i = 1, \dots, n$:

$$\tilde{w}_{M,i} = \frac{N_i P_i}{\sum_{i=1}^n x_{j,i}(1 - w_{j0})} = \frac{N_i P_i}{\sum_{i=1}^n N_i P_i} = \tilde{w}_{M,i}, \quad i = 1, \dots, n$$

Capital Market Line

The efficient set is a single straight line starting from the risk-free point and passing through the market portfolio. This line is called capital market line. Mathematically, capital market line states for any efficient portfolio of assets, its return R satisfies:

$$\mathbb{E}[R] = r_f + \frac{\mathbb{E}[R_M] - r_f}{\sigma_M} \sigma$$

where $\mathbb{E}[R]$ and σ are the mean and standard deviation of R , respectively, and $\mathbb{E}[R_M]$ and σ_M are the mean and standard deviation of the return of the market portfolio.

A short proof: Efficient portfolio $(w_0, (1 - w_0)\tilde{w}_M)$; its expected return is:

$$\mathbb{E}[R] = w_0 r_f + (1 - w_0) \mathbb{E}[R_M] = r_f + (\mathbb{E}[R_M] - r_f)(1 - w_0)$$

and its deviation is:

$$\sigma = |1 - w_0| \sigma_M$$

Thus, we obtain the capital market line (assuming $\gamma_M \geq r_f$). Capital market line is the efficient frontier.

Derivation of Asset Pricing

Consider asset/portfolio i with return R_i , and denote its expectation and standard deviation as $\mathbb{E}[R_i]$ and σ_i , respectively. Consider the mean-variance problem with $z = \mathbb{E}[R_i]$. The optimal portfolio is:

$$\left[\frac{w_0}{\mathbb{E}[R_M] - r_f} \right] \left[(1 - w_0) \tilde{w}_M \right]$$

with $1 - w_0 = \frac{\mathbb{E}[R_i] - r_f}{\mathbb{E}[R_M] - r_f}$.

For any $\epsilon \in \mathbb{R}$, consider the portfolio which is mixture of $(1 - \epsilon)$ of the optimal portfolio and ϵ of asset i . This new portfolio is feasible with expected return z and its standard deviation:

$$\begin{aligned} & \text{Var}((1 - \epsilon)w_0 r_f + (1 - \epsilon)(1 - w_0)R_M + \epsilon R_i) \\ &= (1 - \epsilon)^2 (1 - w_0)^2 \text{Var}(R_M) + 2\epsilon(1 - \epsilon)(1 - w_0) \text{cov}(R_i, R_M) + \epsilon^2 \text{Var}(R_i) \\ &= f(\epsilon) \end{aligned}$$

0 is the minimum point of $f(\cdot)$ because $(w_0; (1 - w_0)\tilde{w}_M)$ is in the minimum-variance set.

Capital Asset Pricing Model (CAPM)

First-order condition:

$$0 = f'(0) = -2(1 - w_0)^2 \text{Var}(R_M) + 2(1 - w_0) \text{cov}(R_i, R_M)$$

leads to:

$$\frac{1 - w_0}{\text{Var}(R_M)} = \frac{\text{cov}(R_i, R_M)}{\text{Var}(R_M)}$$

Capital asset pricing model (CAPM):

$$\mathbb{E}[R_i] - r_f = \beta_i (\mathbb{E}[R_M] - r_f)$$

where

$$\beta_i = \frac{\text{cov}(R_i, R_M)}{\text{Var}(R_M)} = \frac{\mathbb{E}[R_i] - r_f}{\mathbb{E}[R_M] - r_f}$$

draws security market line on $(\beta, \mathbb{E}[R])$ plane.

β_i is referred to as the beta of the asset. $\mathbb{E}[R_i] - r_f$ is termed as expected excess rate of return of asset i ; $\mathbb{E}[R_M] - r_f$ is the expected excess rate of return of market portfolio.

CAPM says expected excess return of an asset is proportional to expected excess return of market portfolio. The proportion is its beta (β_i).

$\beta_i > 0$, i.e., positively correlated; positive expected excess return.

$\beta_i = 0$, i.e., uncorrelated; zero expected excess return.

$\beta_i < 0$, i.e., negatively correlated; negative expected excess return.

It is β rather than σ that contributes to expected excess return.

Economic Interpretation of CAPM Consider the portfolio consisting of market portfolio, ϵ of asset i , and $-\epsilon$ of risk-free asset, the last two components being a zero cost trading strategy with P&L $\epsilon(R_i - r_f)$. The variance of the portfolio is:

$$\text{Var}(R_M) + 2\epsilon \text{cov}(R_i, R_M) + \epsilon^2 \text{Var}(R_i)$$

When $\epsilon \ll 1$, second term dominates third term. If $\text{cov}(R_i, R_M) = 0$, asset i does not affect the variance, just like risk-free asset; thus it has zero risk premium.

If $\text{cov}(R_i, R_M) > 0$, asset i increases the variance; thus it has positive risk premium. If $\text{cov}(R_i, R_M) < 0$, asset i reduces the variance; thus it has negative risk premium.

Key insight: risk is the variance of the portfolio return, not the variance of any single asset's return.

Decompose the Risk

For any asset i with the return R_i , define ϵ_i via:

$$R_i - r_f = \beta_i (R_M - r_f) + \epsilon_i$$

By CAPM, $\mathbb{E}[\epsilon_i] = 0$. By the definition of β_i :

$$\begin{aligned} \text{cov}(R_M, \epsilon_i) &= \text{cov}(R_M, R_i - r_f - \beta_i (R_M - r_f)) \\ &= \text{cov}(R_M, R_i - r_f) - \text{cov}(R_M, \beta_i (R_M - r_f)) \\ &= \text{cov}(R_M, R_i) - \beta_i \text{cov}(R_M, R_M) = 0 \end{aligned}$$

The risky return R_i can be decomposed into two parts: the part commoves with market return and the part uncorrelated with market return. As a result:

$$\sigma_i^2 = \text{cov}(\beta_i (R_M - r_f) + \epsilon_i, \beta_i (R_M - r_f) + \epsilon_i) = \beta_i^2 \sigma_M^2 + \text{Var}(\epsilon_i)$$

The first term, $\beta_i^2 \sigma_M^2$ is called **systematic risk**. The second term, $\text{Var}(\epsilon_i)$, is called **nonsystematic, idiosyncratic, or specific risk**. Only the systematic risk contributes to expected return according to CAPM. Any asset or portfolio on capital market line (efficient frontier) has only systematic risk. Any asset or portfolio below capital market line has nonsystematic risk.

CAPM as a Pricing Formula Certainly equivalent pricing formula: The price P of an asset with payoff Q is:

$$P = \frac{1}{1+r_f} \frac{\mathbb{E}[Q] - \frac{\text{cov}(Q, R_M)(\mathbb{E}[R_M] - r_f)}{\text{Var}(R_M)}}{1+r_f + \beta(\mathbb{E}[R_M] - r_f)}$$

Pricing form of CAPM: The price P of an asset with payoff Q (random variables) is:

$$P = \frac{\mathbb{E}[Q]}{1+r_f + \beta(\mathbb{E}[R_M] - r_f)}$$

First formula is used when the correlation of future payoff with market portfolio is known. Second formula is used when β is known a priori.

Example: The Oil Venture

Example 7.6 (The oil venture) Consider the possibility of investing in a share of a certain oil well that will produce a payoff that is random. The expected payoff is \$1,000 and the standard deviation of return is a relatively high 40%. The beta of the asset is $\beta = .6$. The risk-free rate is $r_f = 10\%$, and the expected return of the market portfolio is .17. What is the value of this share of the oil venture, based on CAPM? From the pricing formula, we have:

$$P = \frac{\mathbb{E}[Q]}{1+r_f + \beta(\mathbb{E}[R_M] - r_f)} = \frac{\$1,000}{1+0.1+0.6 \times (0.17-0.1)} = \$876$$

Example: ABC Fund Analysis

Example 7.4 (ABC fund analysis) The ABC mutual fund has a 10-year record of rates of return. We would like to evaluate this fund's performance in terms of mean-variance portfolio theory and the CAPM.

Step 1 Estimate $\mu := \mathbb{E}[R]$, r_f , $\mu_M := \mathbb{E}[R_M]$, and β from the data:

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n R^i, \quad \hat{\mu}_M = \frac{1}{n} \sum_{i=1}^n R_M^i, \quad \hat{r}_f = \frac{1}{n} \sum_{i=1}^n r_f^i$$

$$\hat{\text{cov}}(R, R_M) = \frac{1}{n-1} \sum_{i=1}^n (R^i - \hat{\mu})(R_M^i - \hat{\mu}_M)$$

$$\begin{aligned} \text{Var}(R_M) &= \frac{1}{n-1} \sum_{i=1}^n (R_M^i - \hat{\mu}_M)^2 \\ \hat{\beta} &= \frac{\text{cov}(R, R_M)}{\text{Var}(R_M)} \end{aligned}$$

Step 2 We write the formula:

$$\hat{\mu} - \hat{r}_f = J + \hat{\beta}(\hat{\mu}_M - \hat{r}_f)$$

If ABC fund is priced according to CAPM, J should be zero statistically. J is known as Jensen's index or alpha (α). For ABC fund, $J > 0$. If you are a CAPM believer, you may think ABC fund is underpriced and thus a good asset. If you are market efficiency believer, you may think CAPM is invalid.

Step 3 Find the estimate of the Sharpe ratio:

$$S = \frac{\hat{\mu} - \hat{r}_f}{\hat{\sigma}}$$

where

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (R^i - \hat{\mu})^2$$

For ABC fund, $S = 0.436$. For S&P 500, $S = 0.467$. If you are a CAPM believer, you may think ABC fund is not efficient.

Testing CAPM Using Linear Regression Suppose there are n risky assets in the market with return rates in the following period to be $R_{i,t}, i = 1, \dots, n$. Suppose the returns of the risk-free asset and of the market portfolio in the following period are $r_{f,t}$ and $R_{M,t}$, respectively.

Decomposition:

$$R_{i,t} - r_{f,t} = \alpha_i + \beta_i (R_{M,t} - r_{f,t}) + \epsilon_{i,t}, \quad i = 1, \dots, n$$

where $\epsilon_{i,t}$ are independent of each other and of $R_{M,t}$.

Vector form:

$$\mathbf{R} - r_f \mathbf{1}_n = \boldsymbol{\alpha} + \boldsymbol{\beta} (R_M - r_f) + \boldsymbol{\epsilon}$$

where $\mathbf{R} := (R_1, \dots, R_n)^\top$, $\boldsymbol{\alpha} := (\alpha_1, \dots, \alpha_n)^\top$, $\boldsymbol{\beta} := (\beta_1, \dots, \beta_n)^\top$, and $\boldsymbol{\epsilon} := (\epsilon_1, \dots, \epsilon_n)^\top$. $\boldsymbol{\epsilon}$ is independent of R_M .

CAPM holds if and only if $\boldsymbol{\alpha} = 0$. To test whether CAPM holds, suppose we have historical data of the asset returns: $R_t, R_{M,t}, r_{f,t}, t = 1, \dots, T$. Suppose the asset returns in different periods are i.i.d. Run vector linear regression to estimate $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$. Reject CAPM if the estimate of $\boldsymbol{\alpha}$ is significantly different from zero.

7.2 Arbitrage Pricing Theory

A Multi-Factor Model

Fix one period and consider n zero-cost strategies in this period with payoffs $f_1, \dots, f_n$, respectively. Assume the following factor model:

$$f_i = \alpha_i + \sum_{k=1}^m \beta_{ik} f_k + \epsilon_i, \quad i = 1, \dots, n$$

where $f_k, k = 1, \dots, m$ are common factors that are payoffs of m zero-cost strategies. ϵ_i represent idiosyncratic risk, i.e., $\mathbb{E}[\epsilon_i] = 0$, $\text{cov}(\epsilon_i, f_k) = 0$, and $\text{cov}(\epsilon_i, \epsilon_j) = 0, i = 1, \dots, n, j = 1, \dots, n, k = 1, \dots, m$, and $i \neq j$. β_{ik} is the loading of asset i on factor k . α_i are constants.

Example: CAPM ($m = 1$). $f = R_M - r_f$, $f_i = R_i - r_f$, $\alpha_i = 0$.

Example: Fama-French three-factor model ($m = 3$). $f_1 = R_M - r_f$, f_2 , usually denoted as SMB, is the difference in the returns of two portfolios that are related to the so-called size premium. f_3 , usually denoted as HML, is the difference in the returns of two portfolios that are related to the so-called value premium.

Vector form of the factor model:

$$\mathbf{f} := (f_1, \dots, f_m)^\top, \boldsymbol{\beta} := \{\beta_{ij}\}, \boldsymbol{\epsilon} := (\epsilon_1, \dots, \epsilon_n)^\top, \boldsymbol{\alpha} := (\alpha_1, \dots, \alpha_n)^\top.$$

Arbitrage Pricing Theory (APT)

Theorem (Arbitrage Pricing Theory (APT): In consequence, there is no arbitrage opportunity in the market. Then, $\boldsymbol{\alpha} = 0$. In addition,

$$\frac{\mathbb{E}[f_i]}{\mathbb{E}[f_j]} = \beta_i \beta_j$$

$\mathbb{E}[f_i]$ is the risk premium of the common factors. The risk premium of asset i is its loading on the common factors multiplied by the risk premium of the common factors.

In CAPM, the only factor is the excess return of the market portfolio, $R_M - r_f$. The risk premium of the common factor is the expected excess return of the market portfolio, $\mathbb{E}[R_M] - r_f$. The loading of asset i on the common factor is its beta.

Statistical Testing of APT

Suppose we have historical data of the payoffs of the n zero-cost strategies: $\tilde{r}_t, t = 1, \dots, T$ and the historical data of the payoffs of the m factors: $f_{i,t}, t = 1, \dots, T$. Suppose the payoffs are i.i.d. across periods. Run multiple linear regression to estimate $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$. Reject APT if the estimate of $\boldsymbol{\alpha}$ is significantly different from zero.

8 Multi-Period Investment Models

8.1 Multi-Period Investment

There are N periods of investment. Denote the beginning and end of j -th period as t_{j-1} and $t_j, j = 1, \dots, N$. There is a risk-free asset and m risky stocks in the market. The net return rate of investing in the risk-free asset in the j -th period is a constant $r_{f,j}$. The excess return rate of investing in stock i in period j is $R_{i,j}$.

Assume that $R_{i,j} := (R_{i,1}, \dots, R_{i,n})^\top, j = 1, \dots, n$ are independent.

Consider an agent with initial wealth x_0 at the beginning of first period. The agent's decision is the allocation of his money in each period.

8.2 Wealth Dynamics

Denote as X_t the wealth of the agent at time $t, j = 0, 1, \dots, N$. Suppose that except for the initial investment, the agent does not inject new money nor withdraw money from the investment account before the terminal time $T := t_N$.

At the beginning of each period j , the agent decides to invest $\pi_{j,i-1}$ dollars in stock $i, i = 1, \dots, m$, and the remaining amount of money in the risk-free asset. Wealth dynamics:

$$X_j = (1 + r_{f,j})X_{j-1} + \sum_{i=1}^m \pi_{j,i-1} R_{i,j}, \quad j = 1, \dots, N$$

8.3 Expected Utility Maximization

Assume that the agent maximizes the expected utility of her terminal wealth,

$$\max_{\pi_{j,i-1}, j=1, \dots, N} \mathbb{E}[u(X_N)]$$

where u is the agent's utility function and $\pi_{j-1} = (\pi_{j,1-1}, \dots, \pi_{j,m,j-1})^\top, j = 1, \dots, N$ are the agent's portfolio strategy.

8.4 Dynamic Programming

We apply dynamic programming to solve the problem. Principle of Optimality: Starting from any point on the optimal path, the remaining decisions must constitute an optimal policy with regard to the point.

9.2 Resolution Parameters

In general, an order $x = (p_x, \omega_x, t_x)$ submitted at time t_x with price p_x and size $\omega_x > 0$ (respectively, $\omega_x < 0$) is a commitment to sell (respectively, buy) up to $|\omega_x|$ units of the traded at a price no less than (respectively, no greater than) p_x .

The lot size σ of an LOB is the smallest amount of the asset that can be traded within it. All orders must arrive with a size $\omega_k \in \{\pm k\sigma : k = 1, 2, \dots\}$. The tick size π of an LOB is the smallest permissible price interval between different orders within it. All orders must arrive with a price that is specified to the accuracy of π .

The lot size σ and tick size π of an LOB are collectively called its resolution parameters.

9.3 Limit Order Book

An LOB $\mathcal{L}(t)$ is the set of all active orders in a market at time t . The active orders in an LOB $\mathcal{L}(t)$ can be partitioned into the set of active buy orders $\mathcal{B}(t)$, for which $\omega_k < 0$, and the set of active sell orders $\mathcal{S}(t)$, for which $\omega_k > 0$.

9.4 Depth

The bid-side depth available at price p and at time t is

$$n^b(p, t) := \sum_{\{x \in \mathcal{B}(t) : p_x = p\}} \omega_x$$

The ask-side depth available at price p and at time t , denoted $n^a(p, t)$, is defined similarly using $\mathcal{S}(t)$.

The bid price at time t is the highest stated price among all active buy orders at time t , i.e.,

$$b(t) := \max_{x \in \mathcal{B}(t)} p_x$$

The ask price at time t is the lowest stated price among all active sell orders at time t , i.e.,

$$a(t) := \min_{x \in \mathcal{S}(t)} p_x$$

9.5 Relative Price

The bid-ask spread at time t is $s(t) := a(t) - b(t)$. The mid price at time t is $m(t) := [a(t) + b(t)]/2$.

For a given price p , the bid-relative price is $\delta^b(p) := b(t) - p$ and the ask-relative price is $\delta^a(p) := p - a(t)$.

For a given order $x = (p_x, \omega_x, t_x)$, the relative price of the order is $\delta_x := \delta^b(p_x)$ if the order is a buy order, and $\delta_x := \delta^a(p_x)$ if the order is a sell order.

The bid-side depth available at relative price δ and at time t is

$$N^b(\delta, t) = \sum_{\{x \in \mathcal{B}(t) : \delta_x = \delta\}} \omega_x$$

The ask-side is defined similarly.

9.6 Price changes in LOBs

Consider a buy (respectively, sell) order $x = (p_x, \omega_x, t_x)$ that arrives immediately after time t . If $p_x \leq b(t)$ (respectively, $p_x \geq a(t)$), then x is a limit order that becomes active upon arrival. It does not cause $b(t)$ or $a(t)$ to change.

If $b(t) < p_x < a(t)$, then x is a limit order that becomes active upon arrival. Upon arrival, $b(t_x) = p_x$ (respectively, $a(t_x) = p_x$).

If $p_x \geq a(t)$ (respectively, $p_x \leq b(t)$), then x is a market order that immediately matches to one or more active sell (respectively, buy) orders upon arrival. Whenever such a matching occurs, it does so at the price of the active order, which is not necessarily equal to the price of the incoming order. Whether or not such a matching causes $a(t)$ (respectively, $b(t)$) to change at time t_x depends on $n^a(a(t), t)$ (respectively, $n^b(b(t), t)$) and ω_x .

In particular, the new bid price $b(t_x)$ upon arrival of a sell market order x is

$$b(t_x) = \max(p_x, q), \text{ where } q = \arg\max_{k \geq b(t)} |n^b(k, t)| > |\omega_x|$$

Similarly, the new ask price $a(t_x)$ upon arrival of a buy market order x is

$$a(t_x) = \min(p_x, q), \text{ where } q = \arg\min_{k \leq a(t)} |n^a(k, t)| > |\omega_x|$$

Put another way, the incoming order x matches to the highest priority active order y of opposite type.

10 Sample Final Exam Paper

Question 1

Consider P&L Z with the following distribution:

$$\mathbb{P}(Z = -12) = 0.01, \mathbb{P}(Z = -6) = 0.02, \mathbb{P}(Z = -4) = 0.01, \mathbb{P}(Z = -1) = 0.02 \\ \mathbb{P}(Z = 0) = 0.5, \mathbb{P}(Z = 1) = 0.24, \mathbb{P}(Z = 3) = 0.2$$

Compute the 95%-VaR, the 95% CVaR, the expected loss conditional on the loss being larger than or equal to the 95%-VaR, and the expected loss conditional on the loss being strictly larger than the 95%-VaR.

Solution 1

We compute the CDF of Z , denoted as $F_{-Z}(x)$. Calculation yields:

$$F_{-Z}(x) = \begin{cases} 0, & x < -3, \\ 0.2, & x \in [-3, -1), \\ 0.44, & x \in [-1, 0), \\ 0.94, & x \in [0, 1), \\ 0.96, & x \in [1, 4), \\ 0.97, & x \in [4, 6), \\ 0.99, & x \in [6, 12), \\ 1, & x \geq 12. \end{cases}$$

As a result, we have:

$$VaR_\alpha(Z) = \inf\{x \in \mathbb{R} : \mathbb{P}(-Z \leq x) \geq \alpha\} = \inf\{x \in \mathbb{R} : F_{-Z}(x) \geq \alpha\}$$

Specifically for different α :

$$VaR_\alpha(Z) = \begin{cases} -3, & \alpha \in (0, 0.2], \\ -1, & \alpha \in (0.2, 0.44], \\ 0, & \alpha \in (0.44, 0.94], \\ 1, & \alpha \in (0.94, 0.96], \\ 4, & \alpha \in (0.96, 0.97], \\ 6, & \alpha \in (0.97, 0.99], \\ 12, & \alpha \in (0.99, 1). \end{cases}$$

Consequently, $VaR_{0.95\%}(Z) = 1$.

The 95% CVaR is calculated as:

$$CVaR_{0.95\%}(Z) = \frac{1}{1-0.95} \int_{0.95}^1 VaR_\beta(Z) d\beta \\ = \frac{1}{0.05} \times [1 \times 0.01 + 4 \times 0.01 + 6 \times 0.02 + 12 \times 0.01] = 5.8$$

The expected loss conditional on the loss being larger than or equal to the 95%-VaR ($VaR_{0.95\%} = 1$) is:

$$\frac{\mathbb{E}[-Z | -Z \geq VaR_{0.95\%}(Z)]}{\mathbb{P}[-Z | -Z \geq 1]} = \frac{\mathbb{E}[-Z | -Z \geq 1]}{0.02+0.01+0.02+0.01} \times [1 \times 0.02 + 4 \times 0.01 + 6 \times 0.02 + 12 \times 0.01] = 5$$

The expected loss conditional on the loss being strictly larger than the 95%-VaR is:

$$\frac{\mathbb{E}[-Z | -Z > VaR_{0.95\%}(Z)]}{\frac{1}{0.01+0.02+0.01}} \times [4 \times 0.01 + 6 \times 0.02 + 12 \times 0.01] = 7$$

Question 2

Consider a wealth manager who is looking into a particular portfolio strategy. She found that the historical estimates of the mean and variance of the portfolio's monthly return are 0.7% and 1.8%, respectively. She also estimated the mean and variance of the market portfolio's monthly return to be 0.6% and 1.2%, respectively, and the correlation between the return rates of the portfolio and the market is 0.9. The monthly risk-free return rate is flat at 0.2% in the past.

(a) According to CAPM, is this strategy a good one? Why?

(b) The director of the wealth manager believes that the Fama-French three-factor model instead of CAPM should be used to assess the portfolio strategy. To this end, the wealth manager estimated the loadings of the portfolio for SMB and HML, which turn out to be -0.3 and 0.7, respectively. She also estimated the expected payoffs for the zero-cost strategies associated with SMB and HML to be 0.4% and 0.3%, respectively. Is this strategy a good one under the Fama-French three-factor model? Why?

Solution 2

(a) We compute β_1 first:

$$\beta_1 = \frac{cov(r_f, r_M)}{Var(M)} = \frac{\rho \sigma_f \sigma_M}{\sigma_M^2} = \frac{0.9 \times \sqrt{1.8\%} \times \sqrt{1.2\%}}{1.2\%} = 1.10227$$

As a result, the risk premium of the portfolio strategy implied by CAPM is:

$$r_i - r_f = \beta(r_M - r_f) = 1.10227 \times (0.6\% - 0.2\%) = 0.44091\%$$

This is lower than the estimated risk premium, namely $0.7\% - 0.2\% = 0.5\%$. So under CAPM, this portfolio strategy is good (it offers a positive alpha).

(b) The loadings of the strategy on the market risk factor, SMB, and HML are 1.10227, -0.3, and 0.7, respectively. The risk premiums of these three factors are 0.6%-0.2%, 0.4%, and 0.3%, respectively. Thus, the model risk premium of the strategy under the Fama-French three-factor model is:

$$1.10227 \times (0.6\% - 0.2\%) - 0.3 \times 0.4\% + 0.7 \times 0.3\% = 0.531\%$$

This is higher than the estimated risk premium of 0.5%. Thus, this strategy is a bad one when benchmarked to the Fama-French three-factor model.

Question 3

Suppose that David's total wealth is \$40,000, in which his car is worth \$15,000. Suppose his utility function is $u(x) = 2\sqrt{x}$. There is a 5% probability that his car would be stolen and he will lose \$15,000. There is also a 10% probability that his car would be damaged in an accident and he will have to spend \$10,000 to repair it.

(a) What is the Arrow-Pratt absolute risk aversion coefficient of David?

(b) If he wants to buy a car insurance which can fully cover his loss in both cases, i.e., stolen and damaged, what is the maximum amount of money he is willing to pay for the insurance?

Solution 3

(a) The Arrow-Pratt absolute risk aversion coefficient is:

$$r(x) = -\frac{u''(x)}{u'(x)} = -\frac{2 \times (1/2) \times (-1/2) x^{-3/2}}{2 \times (1/2) x^{-1/2}} = \frac{1}{x} x^{-1}$$

(b) The maximum amount of money, A , David is willing to pay is the amount such that David is indifferent between his wealth if he purchases the insurance contract and his wealth if he does not purchase the contract.

If David purchases the insurance contract, his wealth is $40000 - A$ and the utility of the wealth is $2\sqrt{40000 - A}$.

If David does not purchase the insurance contract, the expected utility of his wealth is:

$$2\sqrt{40000 - 15000} \times 0.05 + 2\sqrt{40000 - 10000} \times 0.10 + 2\sqrt{40000} \times 0.85 = 390.4524$$

Setting the expected utility of the wealth in the case in which David purchases the insurance equal to the case in which he does not purchase the insurance, namely:

$$2\sqrt{40000 - A} = 390.4524$$

We derive $A = 1886.731$.

Question 4

Consider the following Limit Order Book (LOB) for a security:

Bid Side (Depth Available at Price): \$1.99 (Depth: 2), \$1.98 (Depth: 3), \$1.97 (Depth: 1), \$1.96 (Depth: 2), \$1.95 (Depth: 4).

Ask Side (Depth Available at Price): \$2.02 (Depth: 1), \$2.03 (Depth: 2), \$2.04 (Depth: 2), \$2.05 (Depth: 3), \$2.06 (Depth: 4).

Current Bid Price: \$1.99. Current Ask Price: \$2.02. Current Mid Price: \$2.005.

(a) Suppose an order arrives to buy 3 units of the security at price \$2.00. Calculate the mid price immediately after the arrival of the order.

(b) Suppose an order arrives to sell 1 unit of the security at price \$2.02. Calculate the mid price immediately after the arrival of the order.

(c) Suppose an order arrives to buy 3 units of the security at the market price. Calculate the mid price immediately after the arrival of the order.

(d) Suppose cancellation of an existing sell orders of size 1 at price \$2.02 arrives. Calculate the mid price immediately after the arrival of cancellation.

Solution 4

(a) The order is a buy limit order at \$2.00. Since \$2.00 < Ask (\$2.02), it becomes the new best bid. New Bid: \$2.00. Ask remains: \$2.02. Mid Price:

$$\frac{2.02+2.00}{2} = \$2.01$$

(b) The order is a sell limit order at \$2.02. Since \$2.02 > Bid (\$1.99) and \$2.02 is the current Ask, it adds depth to the ask side but does not change the best prices. Bid: \$1.99. Ask: \$2.02. Mid Price remains:

$$\frac{2.02+1.99}{2} = \$2.005$$

(c) A market buy order for 3 units will walk up the book. It consumes the 1 unit at \$2.02. It consumes 2 units at \$2.03. The new best Ask becomes \$2.04. Bid remains \$1.99. Mid Price:

$$\frac{2.04+1.99}{2} = \$2.015$$

(d) Cancellation of 1 unit at \$2.02. The existing depth at \$2.02 was 1. After cancellation, depth at \$2.02 becomes 0. The next available ask is \$2.03. New Ask: \$2.03. Bid remains \$1.99. Mid Price:

$$\frac{2.03+1.99}{2} = \$2.01$$

Question 5

Consider a 3-period mean-variance portfolio selection problem with one risk-free asset and 1 risky stock. Suppose the initial wealth is \$1. Suppose that the risk-free net return in each period is $r_{f,k} = 0.03$. In addition, for $k = 1, \dots, 3$ the expected excess return vector of the stocks is:

$$\mu_k = \mathbb{E}[R_k] = 5\%$$

and variance of the stock excess return rate is:

$$C_k = Var(R_k) = 0.09$$

Suppose the agent's target is to grow the wealth by 6% in each period. Find the optimal portfolio strategy of the agent.

Solution 5

The number of investment periods is $N = 3$. We first compute D_k :

$$D_k = \mathbb{E}[R_k R_k^T] = C_k + \mu_k \mu_k^T = (5\%)^2 + 0.09 = 0.0925$$

$$\mu_k^T D_k^{-1} \mu_k = (5\%)^2 / 0.0925 = 0.027027028, \quad k = 1, \dots, 3$$

Next, we compute:

$$D_k^{-1} \mu_k = 5\% / 0.0925 = 0.54054054$$

$$s_k = \prod_{j=k}^{N-1} (1 + r_{f,j+1}) = 1.03^{3-k}, \quad k = 0, \dots, 2$$

$$\beta_k = \prod_{j=k}^{N-1} (1 - \mu_{j+1}^T D_{j+1}^{-1} \mu_{j+1}) = 0.972972973^{3-k}, \quad k = 0, \dots, 2$$

The target expected final wealth is $z_0 = 1 \times (1 + 0.06)^3$. (The problem says "grow the wealth by 6% in each period", implying $z_0 = (1.06)^3$).

The optimal strategy formula is $\pi_k^* = -A_k x_k + B_k$.

We calculate the coefficients:

$$A_k := (1 + r_{f,k+1}) D_{k+1} \mu_{k+1} = 1.03 \times 0.54054054 = 0.556756757, \quad k = 0, 1, 2$$

$$B_k := A_k s_k^{-1} \frac{z_0 - \beta_0 z_0}{1 - \beta_0}$$

Substituting the values ($x_0 = 1$, $z_0 = (1.06)^3$, $s_0 = 1.03^3$, $\beta_0 = 0.972973$):

$$B_k = 0.556756757 \times \frac{(1.06)^3 - 0.972972973^3 \times 1.03^3 \times 1}{1 - 0.972972973^3} \times 1.03^{k-3} \\ = 1.30187509 \times 1.03^{k-3}$$

Thus, the values for B_k are:

$$B_k \approx \begin{cases} 1.91400139, & k = 0, \\ 1.227142143, & k = 1, \\ 1.263956407, & k = 2. \end{cases}$$

The optimal portfolio strategy is:

$$\pi_k^* = -0.556756757 x_k + B_k, \quad k = 0, 1, 2$$